

Advanced (Habanero)

- Q1.** What is the vertex of the parabola $y = x^2 + 4x + 4$?
(A) $(-2, 0)$
(B) $(-2, 4)$
(C) $(0, 4)$
(D) $(2, 0)$
(E) $(0, 0)$
- Q2.** What is the axis of symmetry of the parabola $y = x^2 - 6x + 8$?
(A) $x = -2$
(B) $x = 2$
(C) $x = 3$
(D) $x = 5$
(E) $x = 8$
- Q3.** What are the x -intercepts of the parabola $y = x^2 + 2x - 3$?
(A) $x = -3, x = 1$
(B) $x = -1, x = 3$
(C) $x = 1, x = -3$
(D) $x = -2, x = 4$
(E) $x = 0, x = 5$
- Q4.** What is the direction of the parabola $y = -x^2 + 2x - 1$?
(A) up
(B) down
(C) left
(D) right
(E) horizontal
- Q5.** What is the minimum value of the parabola $y = x^2 - 4x + 3$?
(A) -1
(B) 0
(C) 1
(D) 3
(E) 5
- Q6.** What is the equation of the parabola with vertex $(2, 3)$ and x -intercept $(0, 0)$?
(A) $y = x^2 - 4x + 4$
(B) $y = x^2 + 2x + 3$
(C) $y = x^2 - 2x - 3$
(D) $y = x^2 + 4x + 3$
(E) $y = -x^2 + 2x + 3$
- Q7.** What is the axis of symmetry of the parabola $y = x^2 + 4x + 4$?
(A) $x = -2$
(B) $x = 2$
(C) $x = 0$
(D) $x = 4$
(E) $x = -4$
- Q8.** What are the y -intercepts of the parabola $y = x^2 - 2x - 3$?
(A) $y = -3$
(B) $y = 3$
(C) $y = 0$
(D) $y = -1$
(E) $y = 1$

Open Ended Questions (FRQ)

- Q9.** Graph the parabola $y = x^2 - 4x + 3$ and identify its vertex, axis of symmetry, x -intercepts, and direction.

- Q10.** Find the equation of the parabola with vertex $(-1, 2)$, axis of symmetry $x = -1$, and y -intercept $(0, 3)$.

Chef's Tips

- Tip 1: Use graph paper to help students visualize the parabolas.
- Tip 2: Encourage students to check their work by plugging in values for x and y .
- Tip 3: Remind students to label their graphs clearly and provide explanations for their answers.